

Python for QA

Complete Installation & Setup

Beginner guide - from download to first use

Objective

By the end of this manual, you should be able to install the tool, validate the installation, perform the first functional use and recognize common errors. Do not skip validation steps: they confirm that each stage actually worked.

Before you start

No extra runtime is required.

Official download and documentation

<https://www.python.org/downloads/>

Attention

Use only test environments, accounts and devices when a tool can capture traffic, run scans, modify infrastructure or access data. In corporate environments, follow your organization policies.

Open the correct terminal

On Windows, use CMD or PowerShell. On macOS and Linux, use Terminal. Run commands exactly as shown and press Enter after each line.

Windows - step by step

Download the official Windows installer and enable the option that makes Python available from the terminal.

```
py -m pip install --upgrade pip
py -m venv .venv
.venv\Scripts\activate
py -m pip install requests
```

macOS - step by step

Download the official macOS installer, or use the package method recommended by Python.

```
python3 --version
python3 -m pip --version
python3 -m venv .venv
source .venv/bin/activate
python3 -m pip install requests
```

Linux - step by step

Use the distribution package manager or the official Python guidance for your distribution.

```
python3 --version
python3 -m pip --version
python3 -m venv .venv
source .venv/bin/activate
```

```
python3 -m pip install requests
```

Validate the installation

1. python --version
2. py --version
3. py -m pip --version

```
python --version
```

```
py --version
```

```
py -m pip --version
```

Commands and configuration

1. py -m pip install --upgrade pip
2. py -m venv .venv
3. .venv\Scripts\activate
4. py -m pip install requests

```
py -m pip install --upgrade pip
```

Installs the specified component/dependency. Wait for completion and confirm a success message.

```
py -m venv .venv
```

Creates an isolated virtual environment; a .venv directory should exist afterward.

```
.venv\Scripts\activate
```

Activates the virtual environment; the prompt normally shows (.venv).

```
py -m pip install requests
```

Installs the specified component/dependency. Wait for completion and confirm a success message.

PATH, environment variables and a new terminal

When an installer adds an executable to PATH or you configure a variable such as JAVA_HOME/ANDROID_HOME, close the old terminal and open a new CMD/PowerShell. Then repeat the version command. If multiple versions are installed, use where <command> on Windows or which <command> on macOS/Linux to identify the executable being used. Do not remove PATH entries without understanding their origin.

Create a safe test folder

For the first tests, use a simple folder with no production files, for example C:\Projects\qa-lab or ~/qa-lab. This makes generated files, virtual environments, logs and configuration easier to identify.

First use / functional test

1. python -c "import requests; print(requests.__version__)"

```
python -c "import requests; print(requests.__version__)"
```

Expected result

A Python 3.x version, a pip version, an active .venv environment and the installed requests version.

Troubleshooting

- Read the terminal error before repeating commands. Check PATH/environment variables, runtime version, permissions, system architecture and whether a new terminal was opened after environment changes.

- Command not found: confirm the installation directory is on PATH and open a new terminal.
- Wrong version: check whether another older runtime or executable appears earlier on PATH.
- Permission denied: use a user-writable project directory and follow the vendor permission guidance; do not blindly run everything as administrator/root.
- Corporate proxy/firewall: configure the package manager/tool according to your organization network policy rather than disabling security controls.

Final checklist

- Official download and documentation
- Windows - step by step / macOS - step by step / Linux - step by step
- Validate the installation
- First use / functional test
- Expected result
- Troubleshooting

Installation validated. Keep this manual for reinstallations, onboarding and troubleshooting.